

Title

Behavioral and hormonal responses to time-restricted versus conventional meal timing in adults with obesity and glucose dysregulation

Abstract

Background: Whether restricting eating to specific hours influences appetite hormones and eating behavior patterns is not well established. **Methods:** 39 overweight/obese adults (mean age 59 years, BMI 35.4) with prediabetes were randomized to 10-hour restricted eating (n=19) or 16-hour conventional pattern (n=20) for 12 weeks with equal caloric intake. Eating behaviors were assessed by questionnaire; fasting leptin, adiponectin, C-reactive protein, cortisol, and postload ghrelin were measured. **Results:** Weight loss was comparable between groups (2.6% restricted vs 2.4% conventional). Both arms showed increased dietary restraint and decreased disinhibition and hunger sensation scores. No significant between-group differences emerged for appetite hormones or inflammatory markers despite mean reductions in leptin, adiponectin changes, and C-reactive protein. **Interpretation:** Modest weight loss occurred in both conditions, with both groups demonstrating improvements in eating behavior scores. However, significant differential effects on appetite regulation or inflammation were not detected. Further investigation in larger populations is needed.